

NEW PRODUCTS

TECHNOLOGY

COMPONENTS & CHIPS

Automotive multiplexer

Diodes Incorporated has introduced the PI3DPX1225Q, a 10Gbps 6:4 active crossbar multiplexer with a linear



ReDriver, designed for automotive applications, such as smart cockpits and rear-seat

entertainment. It supports USB 3.2 and DisplayPort 2.1 signals via a USB Type-C connector with three operating modes. The compact device is housed in a 40-pin W-QFN package with a 4mm x 6mm footprint, operating on a 3.3V supply. It offers ultra-low latency and supports I2C programming.

Diodes Incorporated

<https://www.diodes.com>

SMD high-current inductors

TDK Corporation has expanded its ERU27M series of SMD high-current, flat-wire inductors, designed for



automotive and industrial applications with increasing power demands. These inductors use an alloy powder core

with a soft saturation characteristic and flat-wire helical winding, offering rated currents from 36A to 48A, inductance values of 2.3µH to 8.5µH, and DC resistances between 0.68mΩ and 1.66mΩ. Custom designs are also available.

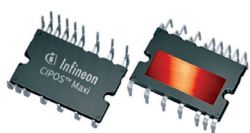
TDK Corporation

<https://www.tdk.com/en/index.html>

Intelligent power module

Infineon Technologies has expanded its

7th generation TRENCHSTOP IGBT7 product line with the CIPOS Maxi intelligent power module (IPM) series,



targeting low-power motor drives. The new

IM12BxxxC1

series, built on 1200 V TRENCHSTOP IGBT7 technology, features the EmCon 7 diode for superior control and efficiency. Designed for HVAC systems, pumps, and motors, it offers EMI protection, thermal management, and a 6-channel SOI gate driver for versatile application use.

Infineon Technologies AG

<https://www.infineon.com>

Sealed tactile switches

Littelfuse, Inc. has updated its C&K Switches KSC2 sealed tactile switch series, now featuring an electrical height enhancement. These surface-



mounted, waterproof switches are ideal for various industries, including medical devices (surgical tools, wearables), transportation (door handles, steering columns), and

high-end consumer products (power tools, lawnmowers). With improved durability, extended lifespan, and clear tactile feedback, the switches ensure reliable, long-term performance. Available in tape and reel format, the series suits industrial applications like elevators and automation systems.

Littelfuse Inc.

<https://www.littelfuse.com>

CryptoAuthentication ICs

Securing key provisioning is critical for protecting sensitive keys from tampering and malicious attacks, especially in consumer, industrial, data centre, and



medical applications. Developing secure key provisioning processes can be complex

and costly. To simplify this, Microchip Technology has expanded its TrustFLEX portfolio with the ECC204, SHA104, and SHA105 CryptoAuthentication ICs. These hardware-based devices offer secure key storage and meet high security standards like JIL and FIPS. They are energy-efficient, compatible with any microprocessor, and ensure confidentiality, integrity, and authenticity across various systems.

Microchip Technology

<https://www.microchip.com>

Coreless current sensor

Molex has launched the percept current sensors, targeting the growing need for accurate busbar current sensing in industrial and automotive sectors.

These sensors utilise Infineon's coreless



current sensing technology and Molex's advanced packaging, resulting

in a compact, lightweight design—86% lighter and up to 50% smaller than competitors. This allows for higher-density device packaging, reducing system weight and offering design flexibility. The sensors maintain 2% accuracy over a broad temperature range, minimising EMI impact with a differential Hall-effect sensor design. Molex's packaging also cuts environmental impact, using